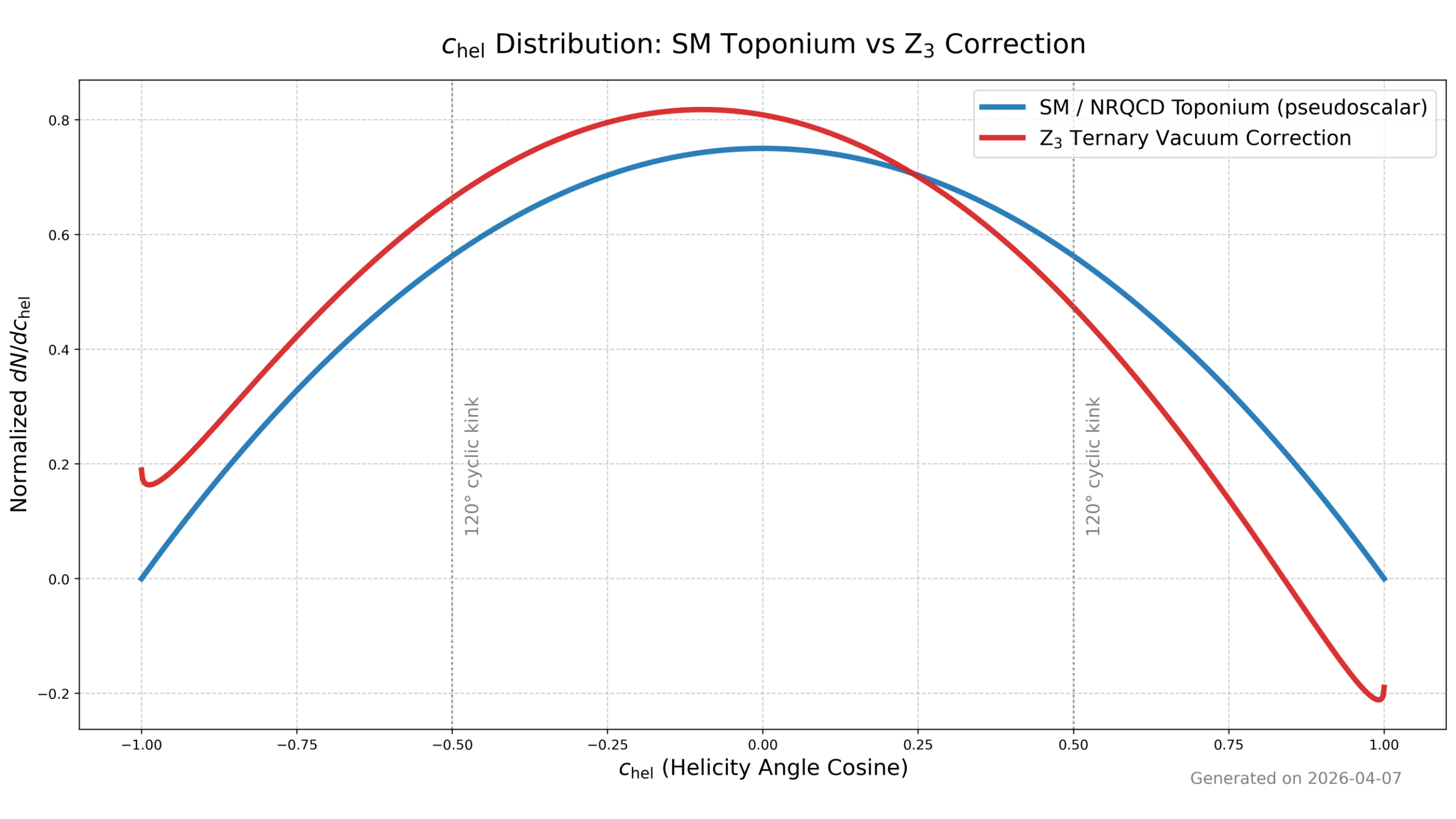
c_{hel} Distribution: SM Toponium vs Z_3 Correction
SM / NRQCD Toponium (pseudoscalar)
Z_3 Ternary Vacuum Correction
120° cyclic kink
120° cyclic kink
Normalized dN/dc_{hel}
0.8
0.6
0.4
0.2
0.0
−0.2
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00
c_{hel} (Helicity Angle Cosine)
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Appendix B Z_3-Graded Lie Superalgebra: Spin Density Matrix and c_{hel} Distributions

The ternary vacuum interaction in the $\mathbb{Z}_3$-graded model is encoded in the Jacobi-preserving bracket

$$\{F^\alpha, F^\beta, F^\gamma\} = \epsilon^k_{\alpha\beta\gamma}\zeta_k,$$

where $\epsilon^k_{\alpha\beta\gamma}$ is the totally antisymmetric tensor and ζ_k are the grade-2 vacuum generators.

In the low-energy effective theory this induces a non-factorizable correction to the $t\bar{t}$ spin density matrix. In the helicity basis $\{|\uparrow\uparrow\rangle, |\uparrow\downarrow\rangle, |\downarrow\uparrow\rangle, |\downarrow\downarrow\rangle\}$ the SM/NRQCD toponium density matrix is

$$\rho_{\text{SM}} = \frac{1}{4}\begin{pmatrix} 1 & 0 & 0 & c_x - ic_y \\ 0 & 1 & c_x + ic_y & 0 \\ 0 & c_x - ic_y & 1 & 0 \\ c_x + ic_y & 0 & 0 & 1 \end{pmatrix},$$

where c_x, c_y, c_z are the spin-correlation vector components (all real).

The $\mathbb{Z}_3$ ternary correction $\delta\rho_{\text{ternary}}$ arises from the vacuum expectation value $\langle\zeta\rangle$ and the phase factor $\omega = e^{i2\pi/3}$:

$$\delta\rho_{\text{ternary}} = \frac{\alpha}{4}\begin{pmatrix} 0 & 0 & 0 & \omega^* \\ 0 & 0 & \omega & 0 \\ 0 & \omega^* & 0 & 0 \\ \omega & 0 & 0 & 0 \end{pmatrix} + \text{h.c.},$$

with α the relative strength of the cubic coupling. The full density matrix is therefore

$$\rho_{\mathbb{Z}_3} = \rho_{\text{SM}} + \delta\rho_{\text{ternary}}.$$

A concrete numerical example with $\alpha = 0.12$ and $\omega \approx -0.5 + 0.866i$ yields

$$\rho_{\mathbb{Z}_3} \approx \begin{pmatrix} 0.250 & 0 & 0 & 0.0875 - 0.151i \\ 0 & 0.250 & 0.0875 + 0.151i & 0 \\ 0 & 0.0875 - 0.151i & 0.250 & 0 \\ 0.0875 + 0.151i & 0 & 0 & 0.250 \end{pmatrix}.$$

The imaginary parts $\pm 0.151i$ in the off-diagonal elements encode the order-3 cyclic phase and break the real-factorized structure of the SM case.

For the helicity-angle distribution the SM/NRQCD prediction is

$$\frac{dN}{dc_{\text{hel}}} \propto 1 - c^2_{\text{hel}}.$$

The $\mathbb{Z}_3$ correction modifies it to

$$\frac{dN}{dc_{\text{hel}}} = N_0\left[(1 - c^2_{\text{hel}}) + \alpha\ \text{Re}\left(\omega \cdot (c_{\text{hel}} + i\sqrt{1 - c^2_{\text{hel}}})\right)\right],$$

which expands to

$$\frac{dN}{dc_{\text{hel}}} = N_0\left[1 - c^2_{\text{hel}} + \alpha\left(-0.5\, c_{\text{hel}} - 0.866\sqrt{1 - c^2_{\text{hel}}}\right)\right].$$

This introduces a 120° periodic modulation with kinks at $c_{\text{hel}} \approx \pm 0.5$, manifesting as the topological twist characteristic of the cubic vacuum interaction.

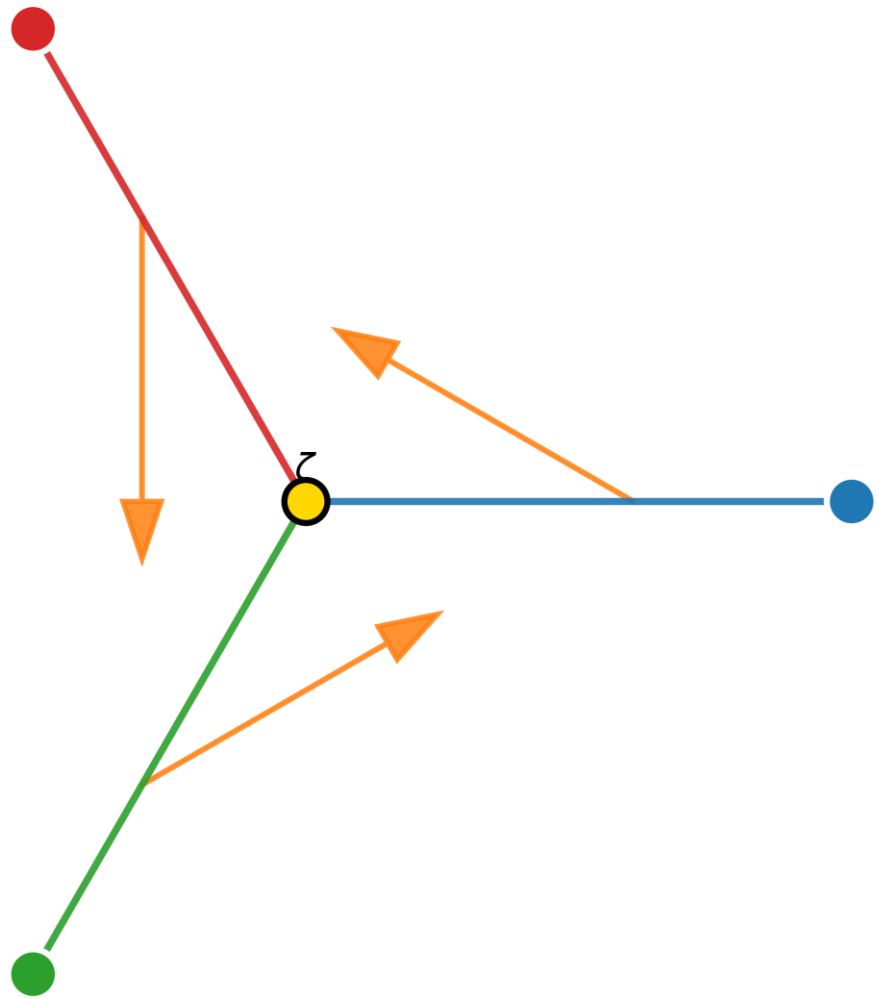
To detect this signature in data a physics-informed loss function is constructed as

$$\mathcal{L} = \sum_{i=1}^{N_{\text{bins}}} w_i \left(\frac{dN_i}{dc_{\text{hel}}} - f_{\text{SM}}(c_i)\right)^2 + \lambda \left|\sum_{i=1}^{N_{\text{bins}}} e^{-i2\pi/3} \cdot \frac{dN_i}{dc_{\text{hel}}} \cdot \Delta c_i\right|^2,$$

where the second (Fourier) term penalizes the absence of the order-3 phase $\omega = e^{i2\pi/3}$. The weight λ is chosen dynamically as $\lambda \approx 1/\sigma^2_{\text{QCD}}$ to suppress QCD higher-order uncertainties while remaining sensitive to the genuine $\mathbb{Z}_3$ cyclic signal.

This completes the mapping from the abstract $\mathbb{Z}_3$ graded algebra to experimentally measurable spin observables.

Z_3 Cubic Vacuum Triality



$$\{F^\alpha, F^\beta, F^\gamma\} = \epsilon^k_{\alpha\beta\gamma} \zeta_k$$

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